microorganisms do not grow. The water requirement of microorganism is defined in terms of the water activity (aw) in the environment. Water activity is defined as the ratio of the water vapour pressure of food substrate to the vapour pressure of pure water at the same temperature. The water activity of most fresh food is above 0.99. The minimum value of aw for the growth of the microorganisms in foods should be around 0.86.

c. Oxidation reduction (O/R) potential

The oxygen tension or partial pressure of oxygen around a food and the O-R potential or reducing and oxidizing power of the food itself influence the type of organisms which can grow and the changes produced in the food. The O-R potential of the food is determined by,

i. The O-R potential of the original food.
ii. The poising capacity, (the resistance to change in potential, of the food.)
iii. The oxygen tension of the atmosphere about the food and
iv. The access which the atmosphere has to the food.

d. Nutrient Content

The kinds and proportions of nutrients in the food are all important in determining what organism is most likely to grow. Consideration must be given to (i) foods for energy (ii) foods for growth and (iii) accessory food substances or vitamins which may be necessary for energy or growth.

e. Antimicrobial constituents

The stability of foods against attack by microorganism is due to the presence of certain naturally occurring substances that have been shown to have antimicrobial activity. Some species contain essential oils that possess antimicrobial activity. Among these are allicin in garlic, eugenol in cloves and cinnamon.

2. Extrinsic factors

These include those properties of the storage environment that affect both the foods and microorganisms present in them. Storage temperature, pH, presence and concentration of gases in the environment are some of the extrinsic factors that affect the growth of microorganisms.



Food Corporation of India: FCI is an organization created and run by Government of India. FCI is a statutory body established through Food Corporation Act, 1964 to meet the following objectives of food policy. Effective price support operations for safeguarding the interests of the farmers.

5.2 Food Spoilage

Spoilage of food can be defined as any visible or invisible change which can make food or product derived from food unfit for human consumption. Spoilage of food not only causes health hazard to the consumer but also causes great economic losses. Spoilage leads to loss of nutrients from food and cause change in original flavor and texture. It is estimated that about 25% of total food produced is spoilt due to microbial activities despite a range of preservation methods available.

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